Lecture Notes (student version)

August 23, 2024

Gravitational-wave Astronomy

Main questions:
what are the main astrophysical sources of GWs,
and what can we learn?

Part 1: BASICS

1. In classical General Relativity, gravitational radiation is generated by time-varying mass-energy quadrupoles and higher moments and currents. We have a gravitational-wave (GW) spectrum [see image].

2. Different types of detectors: resonant mass bars, laser interferometers, pulsar timing, atomic clock timing,… Focus on: **free-mass laser interferometry** (Weiss1983).
   a. See Volume 1 of GWs by Maggiore 2007, and Ch. 11.5 of Gravity by Poisson & Will 2014.

   b. L-shaped detector: A laser source, a beam splitter, two end mirrors mounted on test masses (suspended by small wires).

   c. [see diagram]: the laser beam splits into two orthogonal arms, each bounces off a mirror, and they recombine via destructive interference (ie dark spot on photodetector). If a GW passes through, there will be constructive interference (ie bright spot on photodetector). Thus we detect the GW by the phase difference it induces on the laser beams, which corresponds to detecting variations of the laser beam width on the order of 1/_____th of the diameter of a proton.

3. The amplitude, *ie dimensionless strain*, of a GW is _____, implying that a detector's sensitivity depends on its size.
   a. So, longer detector arms can give smaller strains. But…

   b. The 'noise budget' determines the detector's sensitivity!
      i. low freq. dominated by non-GW noise sources, e.g. seismic noise for ground detectors.
      ii. high freq. dominated by laser shot noise (ground and space based).

   c. Thus, the frequency range of **peak sensitivity depends on L and detector environment.**
   [see aLIGO noise plot]

Part 2: THREE GW REGIMES

Astrophysics -> GW signals: _____(chirps and bursts), _____(sinusoidal), _____(backgrounds/foregrounds).
[see the types of GW signals]
[see the detector sensitivity curves]

1. 10^{-10} Hz to 10^{-7} Hz: _____ [see PTA diagram]
   a. Signals: chirps (massive BBH mergers), continuous (massive BBH inspirals), backgrounds, etc.
   b. Current: IPTA, NANOGrav, EIPTA, CPTA, ParkesPTA, etc.
   c. Future: SKA, etc.
   d. Ex: 2023, all PTAs **detected a stochastic background GW** (Antoniadis et al.; Reardon et al.; Agazie et al.; Xu et al.).

      1) _____ curve, the "smoking gun" prediction from GR. [see NANOGrav plot]

2) Took ~ 20 yrs of observations, but its *source* is still uncertain:

- Expected: _____ (Phinney2001)

[see galaxy merger diagram/gif]

- Others: inflationary GWs, scalar-induced GWs, ultralight dark matter, decaying cosmic strings, etc.

2. 10^{-4} Hz to 10^{-1} Hz: _____ [see the detector sensitivity curves]
   a. Signals: chirps (supermassive BBH mergers), continuous (stellar-mass binary inspirals), backgrounds/foregrounds, etc.
   b. Current: LISA-Pathfinder launched in 2015 - it **exceeded expectations**! See eg Armano+2024
   c. Future: LISA, TianQin, etc.

3. 1 Hz to 10^{3} Hz: _____
   a. Signals: _____, continuous (isolated neutron stars), bursts (supernovae), etc.
   b. Current: _____ network (+LIGO India *soon*) [see LIGO detection volume]

      1) So far, at end of O4a run, we have ~ 200 BBHs, and a few BNSs, and BHNSs.
         a) *GW150914*: the first _____ merger, direct confirmation of GWs and **Kerr Hypothesis.**
         b) *GW170817*: the first _____merger, largest multi-wavelength EM campaign in human history to observe the EM counterparts in X-rays, optical, gamma-rays, etc.
         c) *GW190521*: the first direct evidence of an _____.
         d) *GW200105*, *GW200115* (and possibly *GW190917*, *GW190426_152155*): the first _____!

   c. Future: _____, _____ [see CE detection volume]

   d. Ex: **a black-hole binary merger**
      Waveform model:

      1) _____: the *annus mirabilis* Pretorius2005, Campenalli+2006, Baker+2006

      2) Approximate theory:
         a) post-_____/post-Minkowskian (Blanchet2024 & Damour2018): flat spacetime background, expansion in G and c / expansion in G -> "relaxed Einstein equations".

         b) Effective one-body (Damour2001): ADM-Hamiltonian treatment (inspired by quantum dynamics).

         c) _____ (Thorne1980): radiation theory.

         d) _____(Destounis+2023): curved spacetime background (ie, Weyl scalars, Teukolsky eqtn) -> _____ (Berti+2006,2009), EMRIs (Barack & Pound 2019).

      3) Full waveform: _____ + _____ + _____

      Single-event Data Analysis:
      [see LIGO data analysis diagram]

_____ + _____ + _____

LIGO data analysis paper (https://arxiv.org/abs/1908.11170). C. Berry's abridged guide:

1) Noise model: _____ eg _____

2) Signal Search: _____ analysis (Allen+2012) [see GW150914 and video]

3) Parameter Estimation: **Bayesian Inference**, 15 parameters for 1 binary
   a) Bayes' Theorem:

   b) For 10 points in each dimension, we have 10^{15} likelihood evaluations - yikes!

   c) Instead, use _____ such as MCMC, nested sampling. Some codes that implement these: dynesty, emcee, cpnest, etc.

Population Analysis:

1) Main method: _____(Mandel+2019, Vitale+2019)
2) Probe uncertain astrophysics:
   a) progenitor populations: _____ VS _____
   b) **spins carry clues** of astrophysical origin, eg relativistic spin precession
3) Computational limitations need to be addressed for 10,000+ detections from future detectors.

Part 3: FINAL NOTES

1. A note on binary mergers: for binary total mass M,

   a. at leading PN order (ie the Newtonian quadrupole), $f_{\rm GW} \cong 2\, f_{\rm orb}$,

   b. the binary separation as the Schwarzschild radius, $r = 2GM/c^2$,

   c. via Kepler's 3rd law,

   $f_{\rm GW} \cong 2\, f_{\rm orb} \approx$

   Thus, a **higher** mass binary merges at **lower** frequency (implies multi-band studies)

2. _____ high-energy astrophysics, AKA _____

   a. Coincident EM & GW signals should occur for about _____ (Colombo+2022, Abbott+2023, Corsi+2024)

      i. GWs give the source's _____ and _____ to estimate the host galaxies, and then use EMs to discriminate b/w those possible galaxies.

ii. Multiple detectors needed for accurate sky-localization triangulation

a. BNS merger GW170817: [see gif]
   i. Verified the existence of _____

   ii. Verified that BNS mergers are sources of short GRBs.

   iii. Independent probe of degenerate matter, ie NS equation of state, QCD, etc. (eg. Abbott+2019)

b. BBH mergers in accretion disks can produce post-merger GRBs, jets (not observed yet!)

c. Future detectors will give **unprecedented constraints** on:
   i. Cosmology: Hubble parameter via standard sirens, cosmic filaments & shear, etc.
   ii. Dark matter: NS dark matter cores, supermassive BH dark matter spikes, axion clouds, etc.
   iii. Early Universe: inflationary models, primordial black holes, etc.

d. _____, can use EM and GW datasets from different points in the source's evolution, ie EM from progenitor and GW from the later merger - this is analogous to EM multi-wavelength studies.